PROJECT REPORT - 1

Multi-Sensor IoT Healthcare Data Fusion

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Course / Program: B.Tech (CSE)

Institution: Sitare University

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Dataset: MHEALTH Dataset (UCI)

Research Paper : Link

GitHub Repository: Link

1. Objective

I am working on the MHEALTH dataset to prepare it for building a Human Activity Recognition system with learning. This means I really need to understand the data, its structure, how activities are spread out and what patterns show up. I am also looking for issues like class imbalance or overlapping activities that could cause problems later.
This project is important for things like healthcare monitoring, fitness trackers and smart wearables. If we can understand how people move from sensor data, smart systems can react and even predict what's coming next in real time.

By the end I want a cleaned up understood dataset plus some solid insights about how people behave, a clear sense of what obstacles I need to solve and everything ready to jump into deep learning model training.

2. Problem Description

Human Activity Recognition is all about figuring out what someone’s doing from walking to jumping, just by looking at data from sensors like accelerometers and gyroscopes. You will see this kind of tech behind lots of modern fitness trackers and health monitors. The MHEALTH dataset, for example, grabs data from multiple sensors placed on a person’s body, giving researchers a pretty detailed look at how people move.

Background

Human Activity Recognition is about figuring out what someone's doing from walking to jumping by looking at data from sensors like accelerometers and gyroscopes.
The MHEALTH dataset grabs data from sensors placed on a person's body giving researchers a pretty detailed look at how people move.

Wearable devices have really kicked off a surge in activity recognition. It is everywhere now in fitness apps, health tracking systems, smartwatches and even insurance risk scoring.

Real-life Applications

- Fitness and health tracking
- Smartwatches and other wearables
- Scoring insurance risks
- Systems for care and fall detection

Limitations of Existing Systems

- Poor handling of imbalanced data
- Difficulties of activity detection systems in distinguishing activities that are similar
- Difficulties with raw sensor data for feature extraction

3. Dataset and Feature Description

1. **Dataset Overview:**

Total Samples	1,21,5745
Subjects	10
Activities	13
Features	23
Sampling Frequency	50Hz

Table 1: Dataset Overview Table

This table gives a summary of the dataset used in the project. It shows that the dataset is large, has multiple activities, and includes data from different sensors. This makes it suitable for training a reliable activity recognition model.

2. **Sensor-Based Feature Categorization:**

Feature Type	Sensors Used	Axes	Description
Accelerometer	Chest, Wrist, Ankle	X,Y,Z	Measures linear motion
Gyroscope	Chest, Wrist, Ankle	X,Y,Z	Measures rotational movement
Magnetometer	Chest	X,Y,Z	Measures orientation
ECG	Chest		Measures heart activity
Derived Features	All sensors		Processed and normalized values

Table 2: Sensor-Based Features Categorization Table

This table explains what each sensor measures and how it helps in understanding human activity. Different sensors capture different types of movement like motion, rotation, and body orientation. Together, they provide a complete view of human behavior.

3. **Feature Importance Table**

Feature Group	What It Captures	Example Activities
Accelerometer	Movement intensity	Running, Walking
Gyroscope	Rotational motion	Climbing stairs
Magnetometer	Body orientation	Standing, Lying
ECG	Physical exertion	Running vs Sitting
Combined Features	Overall behavior pattern	All activities

Table3: Feature Importance Table

This table shows how each type of sensor contributes to identifying activities. For example, accelerometers detect movement while ECG shows physical effort. Combining these features helps the model better distinguish between activities.

4. **Multi-Sensor Data Fusion Diagram:**

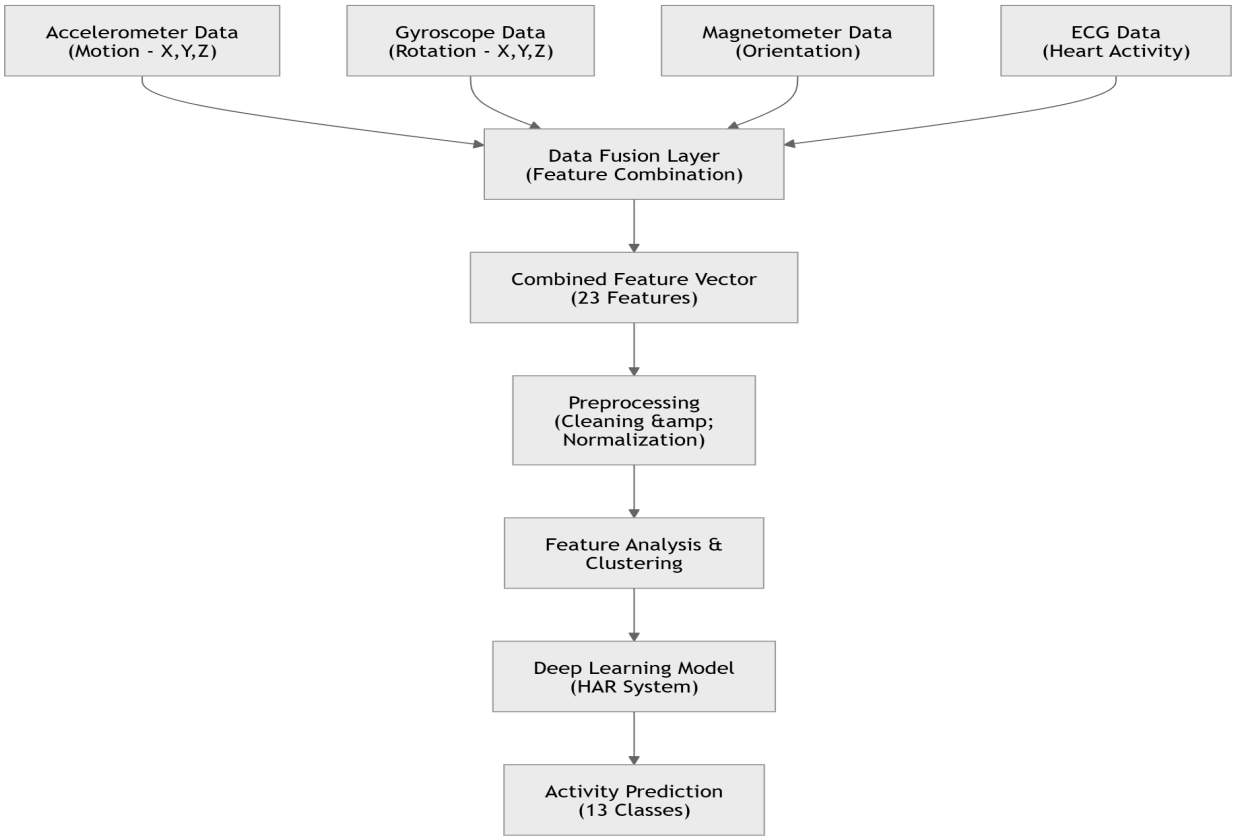

Figure 1: Multi-Sensor Data Fusion Diagram

This diagram shows how data from different sensors is combined into one system. First, sensor data is collected, then processed and merged into a single feature set. Finally, it is used for analysis and prediction of activities.

5. **Explanation of DataFusion:**

The dataset uses multi-sensor data fusion, where signals from different sensors are combined to improve activity recognition.

Single sensor - Limited information

Multi-sensor fusion - Complete understanding of human activity

Types of Fusion Used:

- **Feature-Level Fusion (Early Fusion):** All sensor data is combined before model training.

  Why Fusion is Important:

  - Improves accuracy
  - Reduces ambiguity between similar activities

- Enhances robustness of the model

4. Insights From Research Paper

Concept from Research	What it Means	How it is Used in This Project
Multi-Sensor Data Fusion	Combining multiple sensor signals improves understanding	Accelerometer, Gyroscope, and ECG data are combined
Spatio-Temporal Learning	Data depends on both values and time sequence	Time-window structuring is used
Sequential Data Modeling	Data should be processed as a continuous sequence	Activities analyzed over time, not single points
Data Preprocessing	Cleaning and normalization improve quality	Null class removed, features normalized
Integrated Pipeline	Combining steps improves efficiency	Data cleaning → analysis → clustering → modeling

Table 4: Insights from Research Paper Table

This table connects research concepts with how they are used in this project. It shows how theoretical ideas like data fusion and preprocessing are applied in practice. This helps in building a strong and structured system.

This structured view shows how concepts from the research study are directly applied in this project, improving both understanding and implementation.

Concept Flow from Research to Implementation

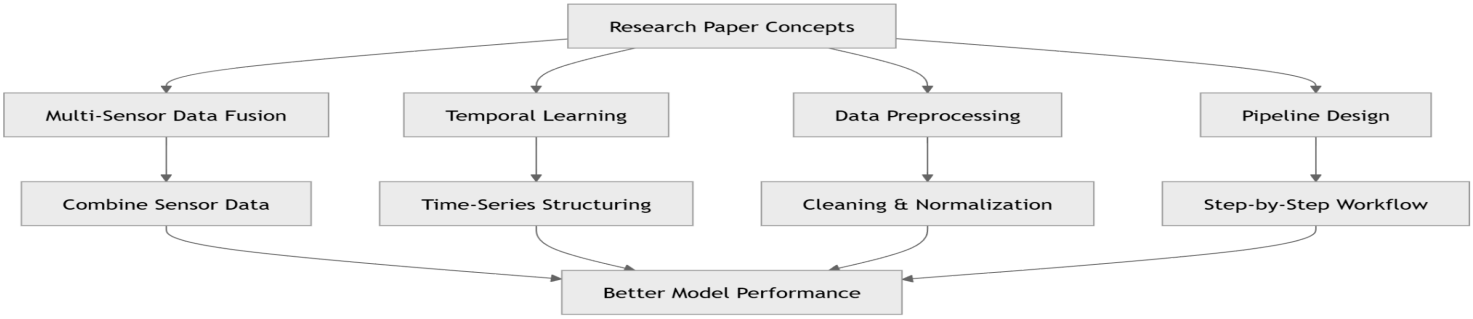

Figure 2: Concept Flow Diagram

This diagram shows how ideas from research are converted into actual implementation. It starts from concepts and moves step-by-step towards building a better model. This makes the workflow easier to understand.

5. Approach / Methodology

This investigation is a structured and systematic procedure used for collecting, understanding and preparing the data set that gives you confidence in making an accurate model prediction.

Procedural steps to achieve this are:

1. Collection of raw sensor data from the data set
2. Data preprocessing and cleaning
3. Exploratory Data Analysis of the raw processed data
4. Analysis of the features (23 overall)
5. Analysis of class distribution
6. K-Means clustering
7. Analysis of patterns in the sensors
8. Extracting insights
9. Data set prepared for model training

Detailed Workflow Pipeline

1. **Raw Data → Cleaned Data:-**

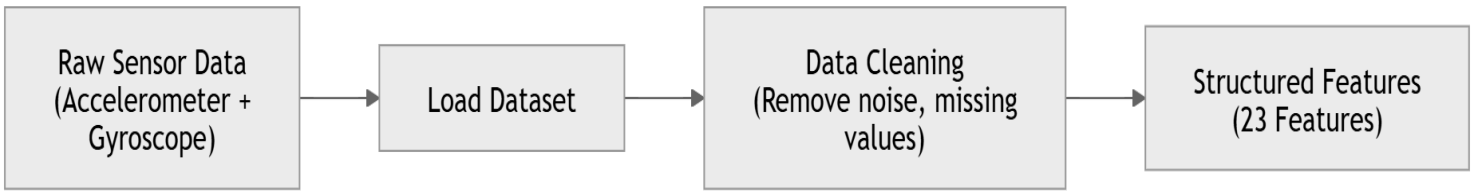

Figure 3: Raw Data to Cleaned Data

This diagram explains how raw sensor data is cleaned and converted into useful features. Noise and missing values are removed to improve data quality. This step is important before applying any analysis or model.

2. **Feature Processing & Labeling:-**

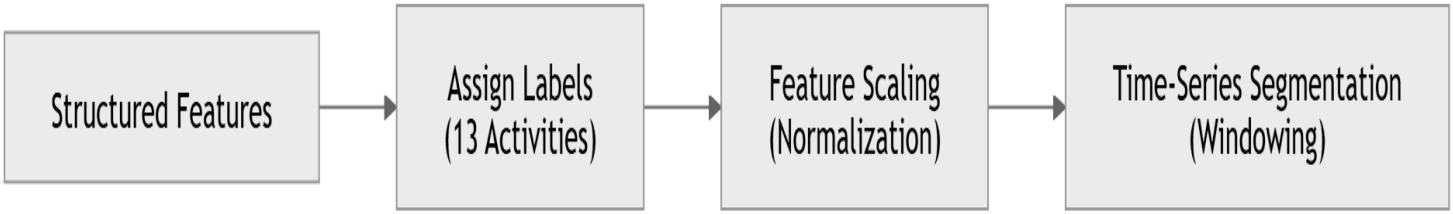

Figure 4: Feature Processing

This diagram shows how features are prepared for training. Labels are assigned, data is normalized, and then divided into time-based sequences. This helps the model understand patterns over time.

3. Data Analysis & Final Dataset:-

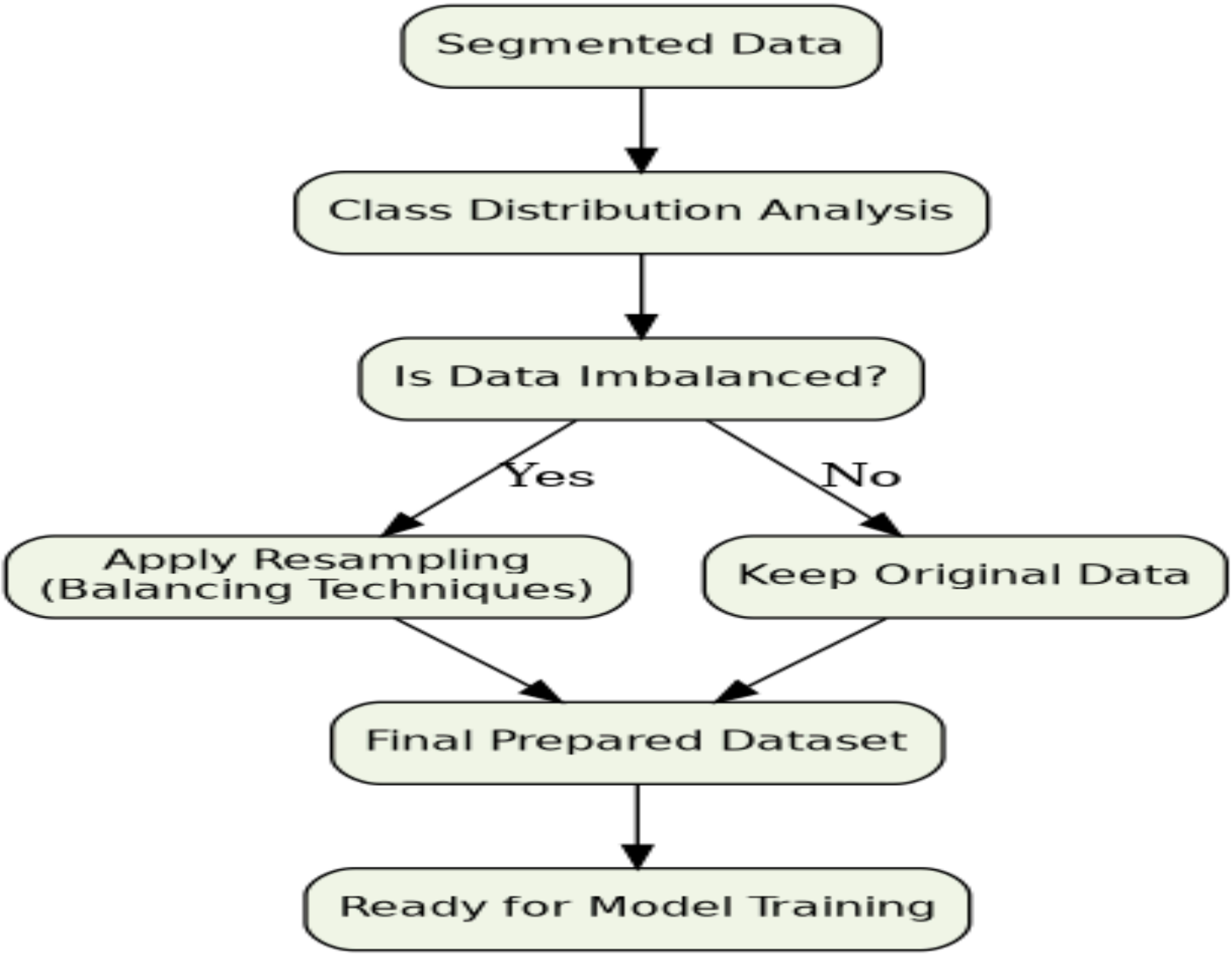

Figure 5: Data Analysis

This diagram explains how the dataset is analyzed step by step. It checks whether the data is balanced or not and applies resampling if needed. Finally, the dataset becomes ready for model training.

Tools and Technologies Used

1. Python
2. NumPy and Pandas
3. Matplotlib
4. Scikit-learn (K-Means)
5. Google Colab and GitHub

6. Data Preparation Algorithm and Logic applied

The primary logic is to convert Raw Sensor Signals to structured Data in a manner that has Useable Meaning:

- Sensor Readings are interpreted from the accelerometer and gyroscope etc. as they are represented as Continuous Values
- Normalization of the Values, allows for each sensor to have an equal contribution to the overall Data Description (Feature Values)
- The Labels will be represented by a number for the purpose of Model Development
- An analysis of any Data Class Imbalance is performed to prevent/classify any learning Bias.

The Techniques that produce Clustering from the Data validate the means of determining whether or not the activity being tracked can be separated from each other based on the Data Characteristics.

Important Code Snippets and Explanation

1. Data cleaning

```
cleaned_dataset = dataset[dataset["Activity"] != 0].reset_index(drop=True)
```

Remove the null class activities from the data.This way, the impact of class imbalance will be reduced and the overall effectiveness of the algorithm will be increased.

2. Feature Normalization

```
scaler = StandardScaler()

Xnormalized = scaler.fit transform(X)
```

The normalization ensures that all of the features are at the same scale. Without it, some features will have too much influence over others.

3. K-Means Clustering

```
kmeans = KMeans(no.  clusters=13, random state=42)
clusters = kmeans.fit predict(Xscaled)
```

K-means aims to cluster the data into groups of similar samples. Since there were 13 activities, the k-means algorithm has been used with 13 clusters to check whether different activities would form clusters.

4. Elbow method logic

```
inertia = []
for k in range(2, 15):
   kmeans = KMeans(no. clusters=k)
```

```
kmeans.fit(Xscaled)
inertia.append(kmeans.inertia)
```

Inertia represents the compactness of clusters. The elbow corresponds to the point when adding further clusters results in only marginal improvements in the clusters' overall configuration. In the current dataset, the elbow point was reached at 13 clusters.

7. Implementation Details

The implementation follows a structured pipeline consisting of multiple modules, each responsible for transforming the data step-by-step.

1. **Modules Implemented:** Each module performs a specific task.

1. Data Preprocessing Module

Data Cleaning
Feature Normalization

2. Exploratory Data Analysis Module

Distribution of Data
Class Imbalance Detection: Null class contributes approximately 71.8% of the total data, indicating a highly imbalanced dataset.

3. Clustering Module

K-Means Clustering: Groups similar data points based on feature similarity.

Elbow Method: Used to determine the optimal number of clusters. The result shows K = 13, which matches the number of activity classes.

4 Sensor Analysis Module

Accelerometer Data: Measures motion intensity. Activities like running Gyroscope Data: Captures rotational movement. Activities like walking show periodic patterns.
Static Activities: Show minimal variation (sitting, standing)
Dynamic Activities:Show strong fluctuations (running, jumping)

5. Final Data Preparation Module

Feature Refinement: Relevant features are selected and noise or redundant data is reduced.
Data Formatting: Inputs - X :- sensor features and Outputs- y:- activity labels.
Train-Test Split: Data is split into training (for learning) and testing.

Time-Series Structuring: Data is organized into fixed-size sequences (time windows) to capture temporal patterns.

8. Output / Results

This part of the report shows what we found out from looking at the MHEALTH dataset. We have got pictures, like graphs and plots, and some examples of what the results look like. We will also explain what each one means, so you can see how well our method works.

1. Dataset Overview Output

```
Total Samples:        1,215,745
Number of Features:   23  (sensor channels; excludes target label)
Number of Activities: 13  (labels 0-12, incl. null class 0)
```

Figure 6: Dataset Summary Output

It highlights that the dataset is large and suitable for training deep learning models. It has 23 features and 13 activities.

2. Activity Class Distribution

Figure 7: Activity Distribution of Dataset

This graph shows how different activities are distributed in the dataset. It is clear that the “Null” activity dominates most of the data. This imbalance can affect model performance if not handled properly.

3. Activity Distribution (Excluding Null Class):

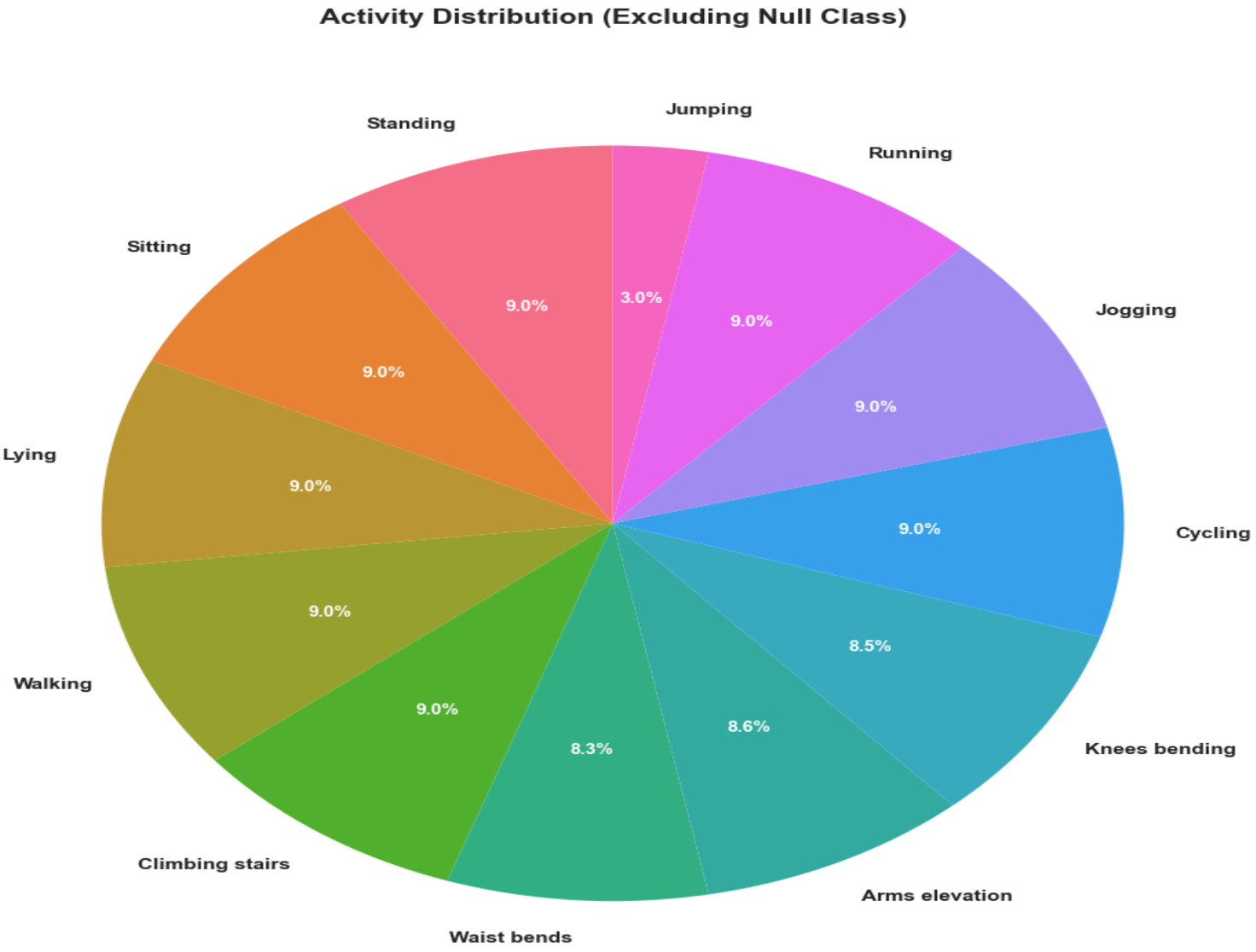

Figure 8: Activity Distribution (Excluding Null Class)

After removing the Null class, the data becomes more balanced across activities. Most activities now have similar representation, which is better for training. However, some activities still have slightly fewer samples.

4. Elbow Method Result:-

Figure 9: Elbow Method for Optimal Clusters

This graph helps in finding the optimal number of clusters for K-Means. The point where the graph bends (elbow) shows the best number of clusters. Here, it is 13, which matches the number of activity classes..

5. Clustering Visualization:-

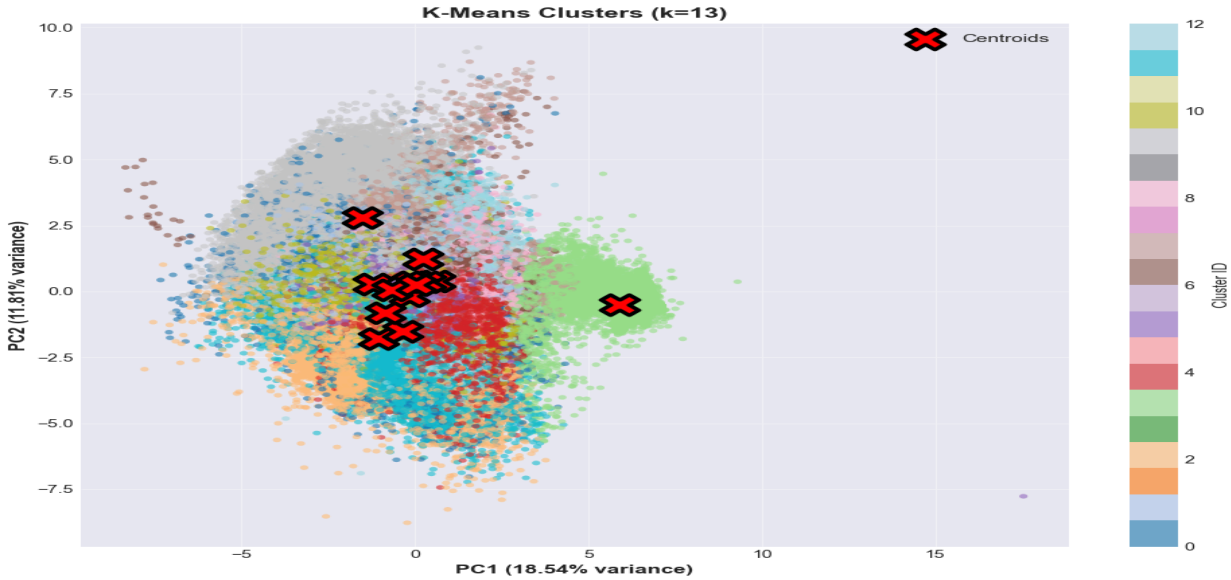

Figure 10: K-Means Clustering Result

This graph shows how data points are grouped into clusters. Different colors represent different clusters of activities. Some clusters are clearly separated, while others overlap slightly.

6. **Clusters Size Distribution:**

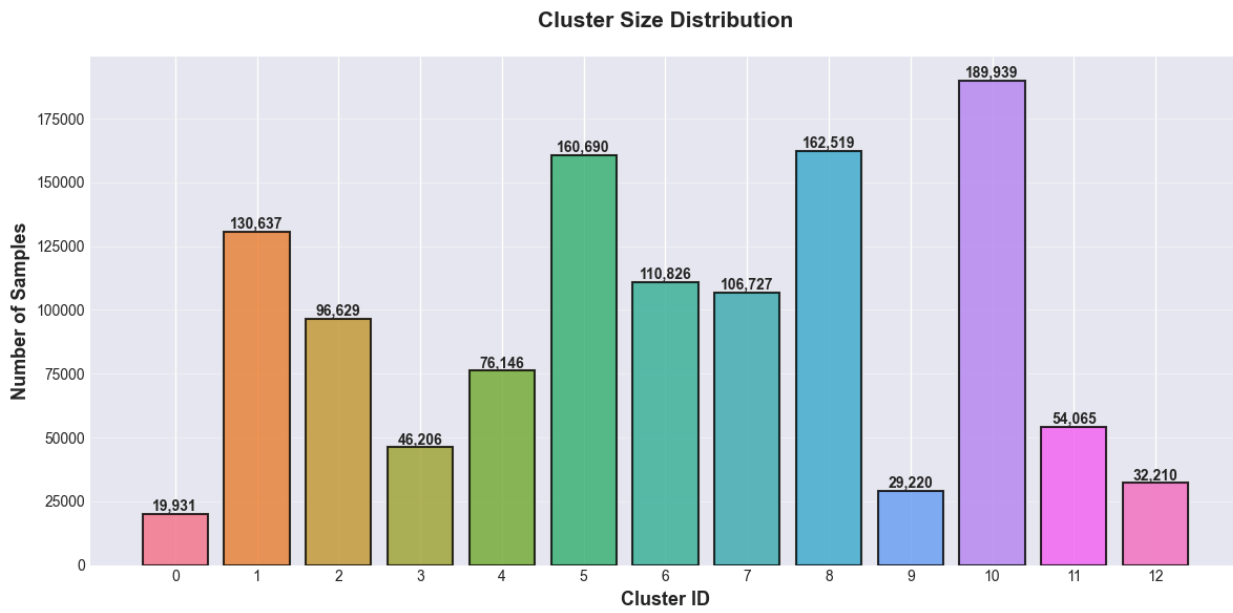

Figure 11: Clusters Size Distribution

This graph shows how many samples are present in each cluster. Some clusters have more data while others have less. This imbalance reflects real world activity patterns and can affect model learning.

9. Key Learning

This part of the project really put theory into practice.

1. **What I Learned**

   I learned that working with the MHEALTH dataset requires data preparation and analysis to build an accurate HAR system

   The insights gained from this project can be applied to real-life applications such, as fitness and health tracking, smartwatches and care systems.

   Diving into those sensor datasets I started to notice the different patterns that various human activities leave on accelerometer and gyroscope readings. Some activities make spikes while others are smooth and steady. It became clear that you can't just throw data into a machine learning model, you need to preprocess it and do a solid analysis.

2. **Skills Developed**
   - Data preprocessing, including cleaning and normalization of 23 sensor features
   - Performing Exploratory Data Analysis (EDA) to understand data distribution
   - Identifying class imbalance (Null class = 71.8%)

- Applying K-Means clustering and using the elbow method (K = 13)
- Interpreting sensor signals for activities like static versus dynamic ones
- Organizing time-series data so it's ready for deep learning models

3. Challenges Faced

- The dataset was large, over 1.2 million samples which made everything slower.
- Dealing with a dominant Null class really messed up the class balance
- Some activities, like sitting and standing looked a lot alike in the data. Were hard to tell apart.
- Sensor signals can get messy. It takes real effort to pull out useful insights

4. Solutions and Improvements

- I used visualization distribution plots and cluster graphs to understand the data.
- I dealt with the class by either removing it or analyzing it separately which helped me see the true activity patterns.
- Running K-Means clustering gave me a way to check if different activities were really separating out the way they should.
- I made sure to normalize the data and structure it into sequences, which gave the models a shot at making accurate predictions.

10. Conclusion

The analysis and preparation of the MHEALTH dataset for Human Activity Recognition are complete. I identified patterns from the sensor data and found clusters that support the claim of meaningful groupings of all activities.
The dominant Null class caused class imbalance challenges during this analysis.
The data was cleaned, normalized and organized in a time series event structure and is now ready for the implementation of learning models in subsequent phases of development.